OpenClaw Core Concepts

OpenClaw Core Concepts
- 2. Course Content
- 3. Core Concepts
  - 3.1 Chat Channels
  - 3.2 Agent
  - 3.3 Sessions and Memory
  - 3.4 Tools
  - 3.5 Skills
  - 3.6 Models
  - 3.7 Gateway Architecture
- 4. Core Concepts Relationship Diagram
- 5. Chapter Summary

2. Course Content

This chapter will systematically explain the seven core concepts of OpenClaw:

1. **Chat Channels** - Platforms for AI to communicate with users
2. **Agent** - The core brain of the AI agent
3. **Sessions and Memory** - Context management capabilities
4. **Tools** - Limbs for executing specific operations
5. **Skills** - Encapsulated functional modules
6. **Models** - The engine that drives AI thinking
7. **Gateway** - The nervous system hub of the system

---

3. Core Concepts

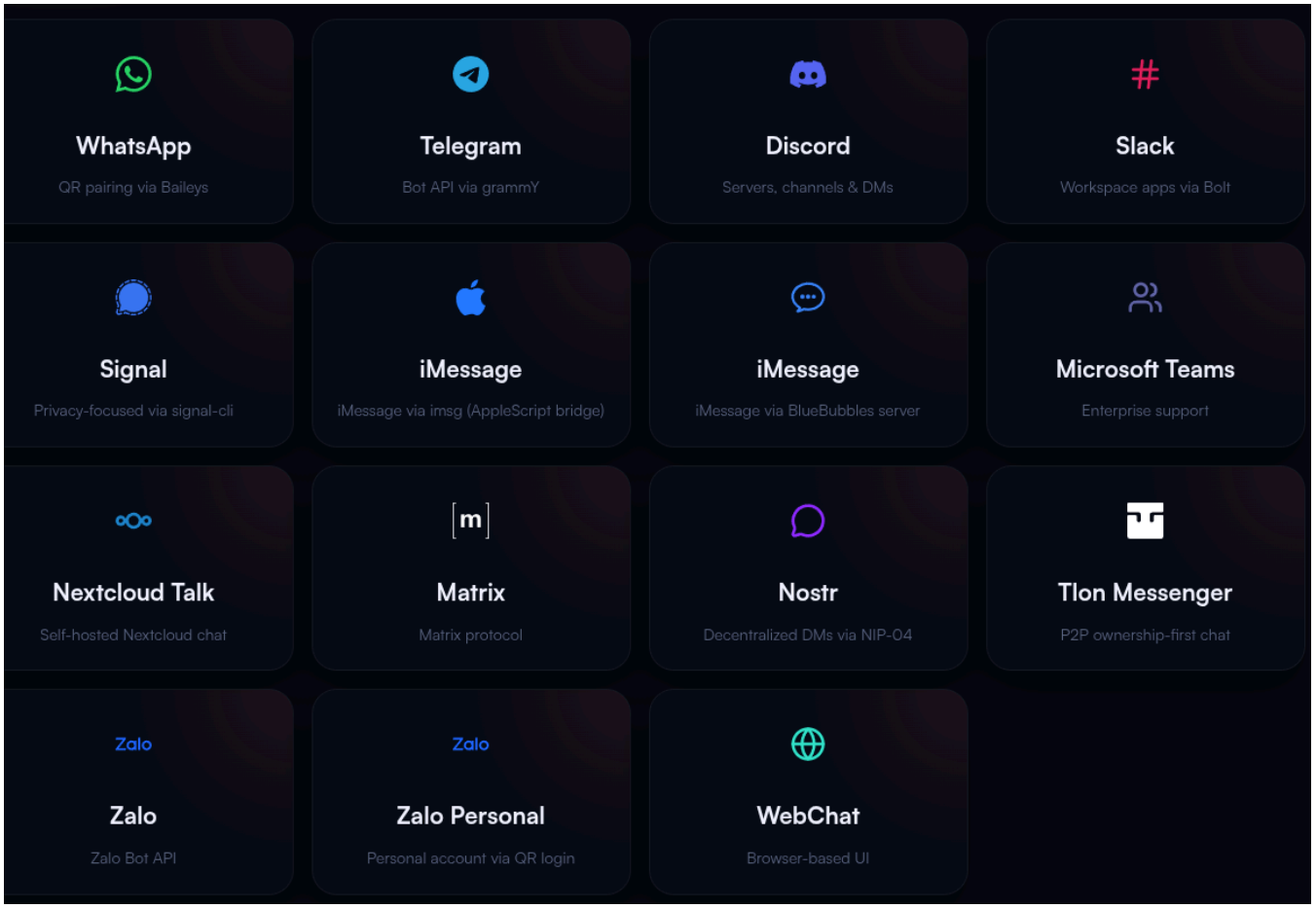
3.1 Chat Channels

What are Chat Channels?

Chat channels are communication platforms for users to interact with AI Agents. They can be understood as the AI employee's "desk" or "workplace".

Supported Channel Types:

Channel Type	Representative Platforms
Instant Messaging	Slack, Discord, Microsoft Teams
Social Apps	Telegram, WhatsApp, Signal
Web Interface	WebChat, REST API



Core Characteristics of Channels:

- **Plug and Play** - Enable required channels without redeployment
- **Many-to-Many Binding** - One channel can connect to multiple Agents, and one Agent can serve multiple channels

---

3.2 Agent

The Agent is the core decision-making unit of OpenClaw. You can think of it as your "virtual employee" or "brain". Each Agent has independent:

- **Persona Prompt** - Defines the AI's behavior style and role positioning
- **Model Selection** - Determines which LLM to use (such as GPT-4, Claude, etc.)
- **Channel Binding** - Specifies which platforms to work on

Key Features:

- **Multi-Agent Parallel** - A single OpenClaw instance can run multiple Agents
- **Role Isolation** - Different Agents do not interfere with each other, with independent configuration

---

3.3 Sessions and Memory

Why is Memory Needed?

The session and memory system allows AI to remember context and provide coherent conversational experiences.

OpenClaw's Three-Tier Memory Architecture:

Short-term Memory

- Storage Format: Daily log files (`memory/yyyy-mm-dd.md`)
- Retention Period: Recent 1-2 days of conversations
- Purpose: Handling context for current tasks

Near-term Memory

- Storage Location: `sessions/` directory
- Management: Sliding window compression, maintaining complete session archives
- Purpose: Multi-round conversation context tracking

Long-term Memory

- Storage Content: User preferences, important decisions, historical experiences
- Loading Method: Automatically loads `memory.md` at the start of each conversation
- Purpose: Achieving personalized intelligence and continuous learning

---

3.4 Tools

What are Tools?

Tools are the "limbs and senses" through which Agents interact with the external world, enabling AI to execute actual operations. If an LLM is the brain, then tools are the limbs attached to that brain.

OpenClaw Built-in Tools:

Tool Type	Function Description	Use Cases
File System	Read/write files, create directories, search content	Document processing, data organization
Network Requests	HTTP API calls, Webhook triggers	Third-party service integration
Shell Commands	Execute system commands, run scripts	Automated operations, batch processing
Scheduled Tasks	Cron scheduling, periodic execution	Scheduled backups, daily report sending

Tool Execution Flow:

```
User Instruction → Agent Understands Intent → Select Appropriate Tool → Execute Operation → Return Result
```

3.5 Skills

Skills vs Tools: What's the Difference?

- **Tools (Tool)** are basic capabilities, such as "open file", "click mouse"
- **Skills** are encapsulated business logic, such as "robotic arm grasping objects", "navigating to a certain location"

Analogy for Skills:

If we compare an Agent to a robot:

- **Agent** = Robot's brain (responsible for thinking and decision-making)
- **Tools** = Robot's limbs (responsible for executing basic actions)
- **Skills** = Robot's training manual (tells the robot how to complete specific work)



Core Characteristics of Skills:

- **Atomicity** - A skill does one thing only (e.g., "grasp object")
- **No Decision-Making** - Skills execute passively, do not judge "whether to do"
- **Standardization** - Follows OpenClaw specifications, can be reused across scenarios
- **Pluggable** - Install/uninstall at any time

Skill Sources:

- **ClawHub** - Community-maintained skill marketplace

- **Custom Skills** - Written by developers (Markdown or TypeScript)

3.6 Models

What are Connected Models?

Models refer to the large language models (LLM) that drive Agent thinking. They are the "intellectual engine" of AI. OpenClaw itself does not train models but serves as a "middleware" connecting various LLM providers.

Model Functions:

- **Understanding Intent** - Parsing natural language user input
- **Task Planning** - Breaking complex goals into executable steps
- **Tool Selection** - Deciding which tool or skill to call
- **Self-Correction** - Adjusting strategy based on execution results

Supported Model Providers:

Provider	Representative Models	Characteristics
OpenAI	GPT-4o, GPT-4 Turbo	Strong comprehensive capability, complete ecosystem
Anthropic	Claude 3.5, Claude Sonnet	Excellent long-text processing, high security
Google	Gemini Pro, Gemini Ultra	Strong multimodal capability
Mistral	Mistral Large, Mixtral	Open-source models, high cost-effectiveness
Ollama	Llama 3, Qwen, etc.	Local deployment, privacy protection
Others	Any OpenAI-compatible interface	Custom model integration

3.7 Gateway Architecture

What is the Gateway?

The Gateway is the "nervous system hub" and "general commander" of OpenClaw. It is a long-running background service that uniformly manages connections across all communication channels and coordinates communication between Agents, clients, and nodes.

Core Responsibilities of the Gateway:

Message Routing

- Receives messages from 50+ channels (WhatsApp, Telegram, Slack, etc.)
- Verifies user identity and permissions
- Distributes to corresponding Agents based on session ID

Session Management

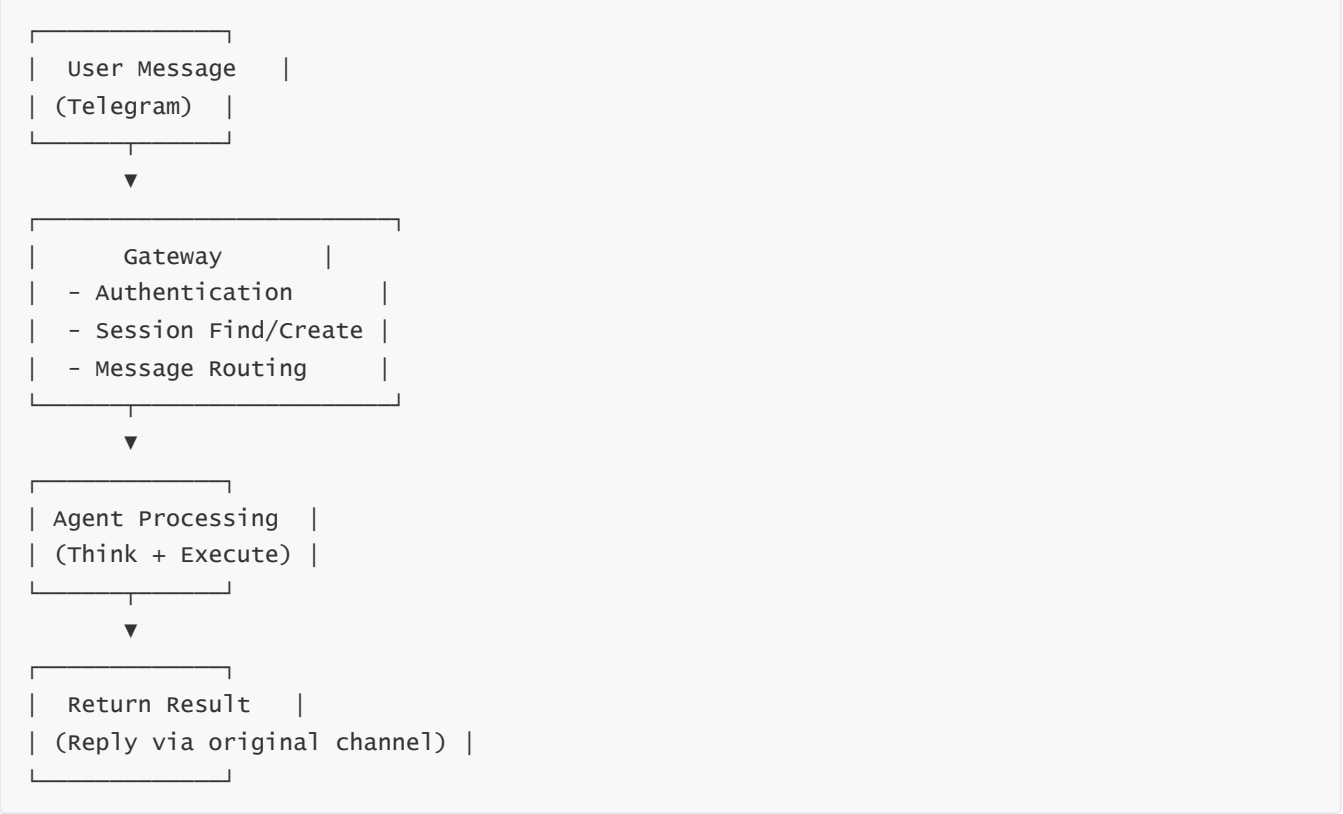
- Maintains the state of all sessions

- Supports "explicit parallelism, default serial" processing
- Heartbeat detection and exception retry

Task Scheduling

- Scheduled task triggers (Cron)
- Load balancing and failover
- Performance monitoring and log recording

Gateway Workflow:



---

4. Core Concepts Relationship Diagram

One-sentence Summary of Each Concept's Role:

Concept	Analogy	Core Function
Chat Channels	Office/Desk	User interaction entry
Agent	Virtual Employee/Brain	Decision and planning
Sessions and Memory	Work Notes/Experience	Context management
Tools	Limbs and Senses	Basic execution capability
Skills	Training Manual	Encapsulated business logic and process

Concept	Analogy	Core Function
Models	Intellectual Engine	Thinking and reasoning
Gateway	Nervous System Hub	Scheduling and coordination

---

5. Chapter Summary

- OpenClaw is a local-first AI Agent operating system, not a simple chatbot
- The role positioning and relationships of the seven core concepts
- The layered architecture design of channels, Agents, skills, and tools
- Supported model providers and selection strategies
- The key role of Gateway as the system hub